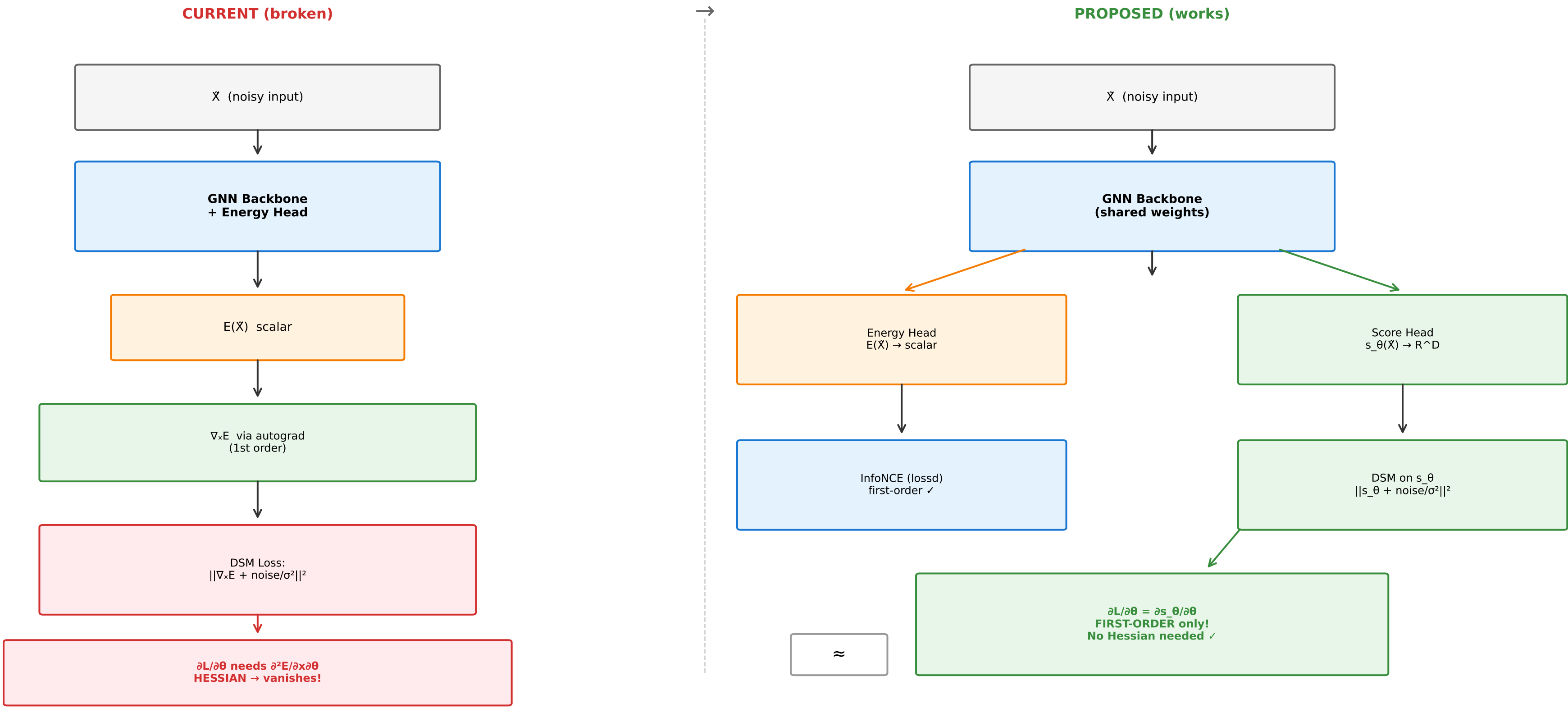
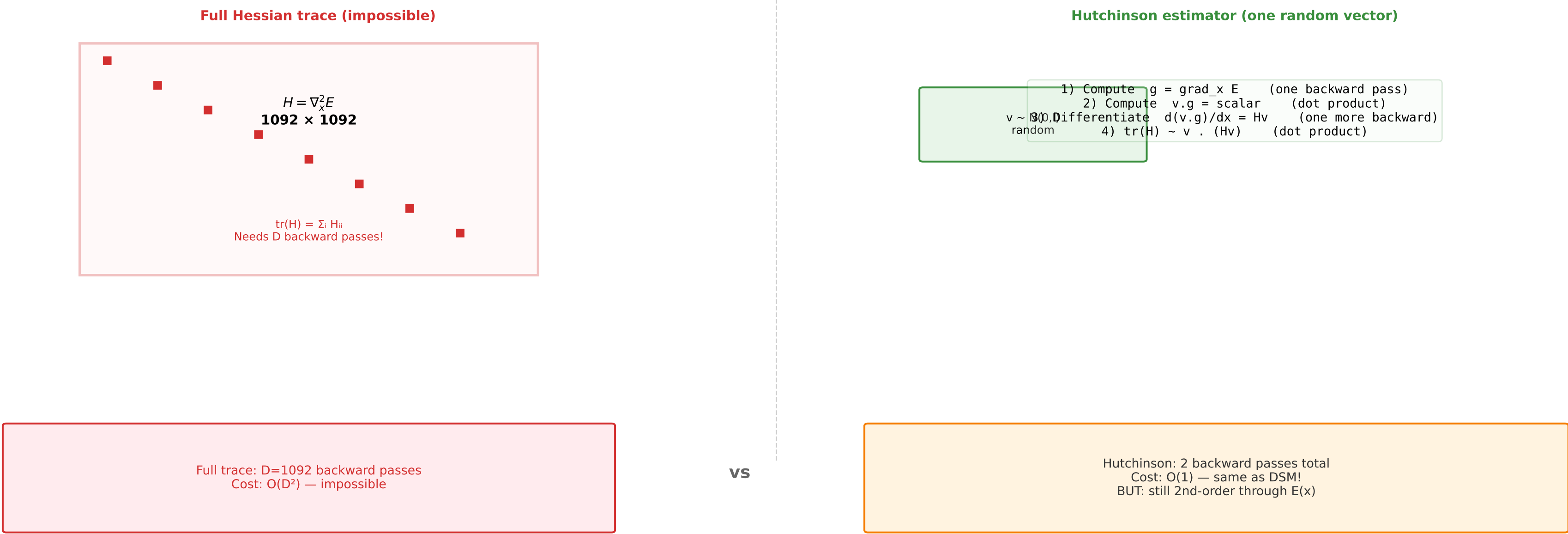


1. Direct Score Prediction Head



2. Fisher Divergence with Hutchinson Trace Estimator

Implicit Score Matching (no noise needed!): $L_{ISM} = \frac{1}{2} ||\nabla_x E||^2 + \text{tr}(\nabla_x^2 E)$



3. Noise Conditional Score Network (NCSN)

User concern: "Doesn't this change the network from predicting energy to predicting score?" $\rightarrow$ YES.

